Exploring the shapes of graphs using technology

Task A: Reviewing plotting graphs of equations

1) Create a table of values to plot the linear graph of $y = 11 - 5x$ for x-values in the range $-2 \leq x \leq 3$ in steps of 0.5

x	$11 - 5x$
-2	21
-1.5	18.5
-1	16
-0.5	13.5
0	11
0.5	8.5
1	6
1.5	3.5
2	1
2.5	-1.5
3	-4

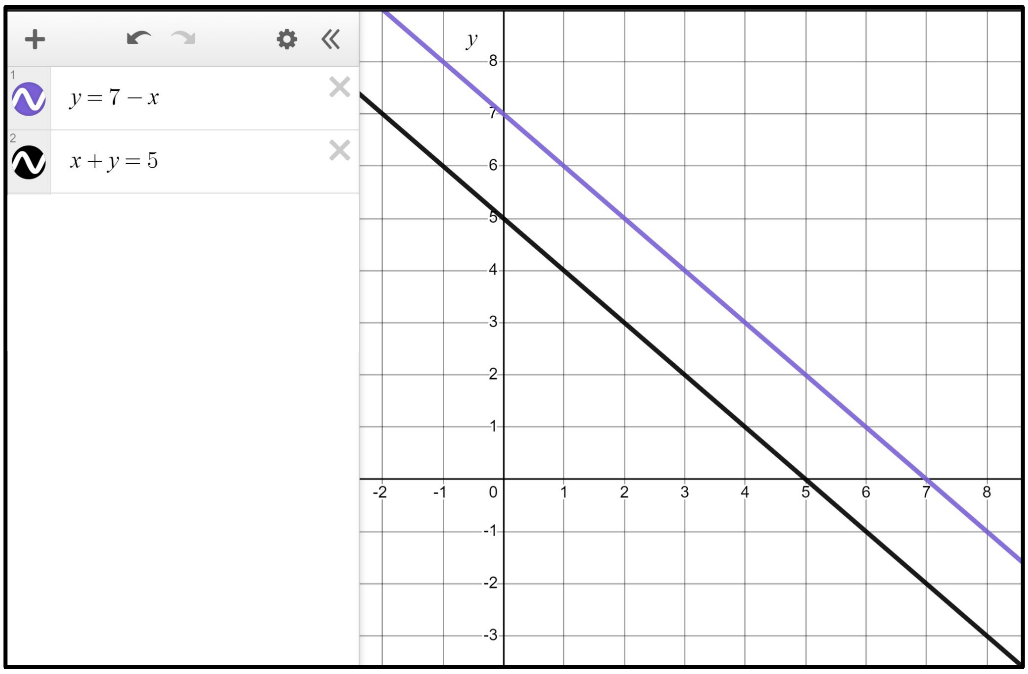
2)

a) Without using a table plot the linear graph of $y = 7 - x$

b) Without using a table plot the linear graph of $x + y = 5$

$x + y - x = 5 - x$

$y = 5 - x$



c) What do you notice about the shape of the two graphs? Write a sentence.

You might have written:
Both are linear graphs. Both have a gradient of -1 making them parallel.

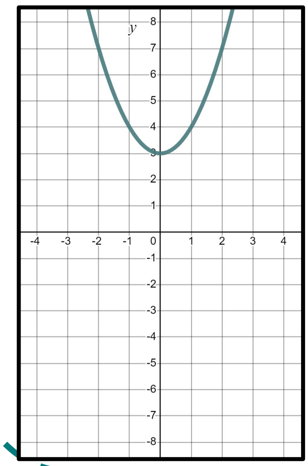
Task B: Plotting quadratic graphs

1) Categorise these equations into **quadratic**, linear, and 'other'.

Quadratic	b) $y = x^2 - 5$	d) $y = x^2 - x + 5$	e) $y = x^2 - 5x$
	f) $y = x^2 - 5x + 1$	h) $y = 5 + x^2$	k) $y = x(x + 2)$
Linear	a) $y = x - 5$	g) $y = 5 + 2x$	j) $y = 5(x + 2)$
'Other'	c) $y = x^4 - 5$	i) $y = x^5 + x^2$	l) $y = x(x^2 + 2)$

2) Use technology to plot the graphs of these quadratic **equations**. They all form **parabolas**. Spot the odd one out.

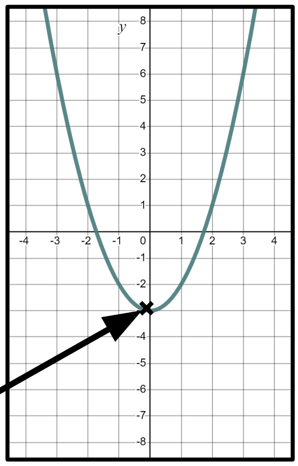
a) $y = x^2 + 3$



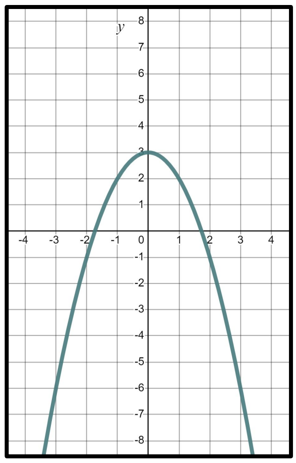
b) $y = 3 + x^2$

a) and b) Addition is commutative, hence the same graph.

c) $y = x^2 - 3$



d) $y = 3 - x^2$



You might have said, "c) is the odd one out because it is the only one with a negative y-intercept".

You might have said,

"d) is the one out. It is upside down".

*Whilst it is very different it is still a **parabola**. It looks upside down but we would tend to say it "Opens downwards" or is '∩-shaped'. Importantly, we still have a **quadratic** equation and a **parabola**.*